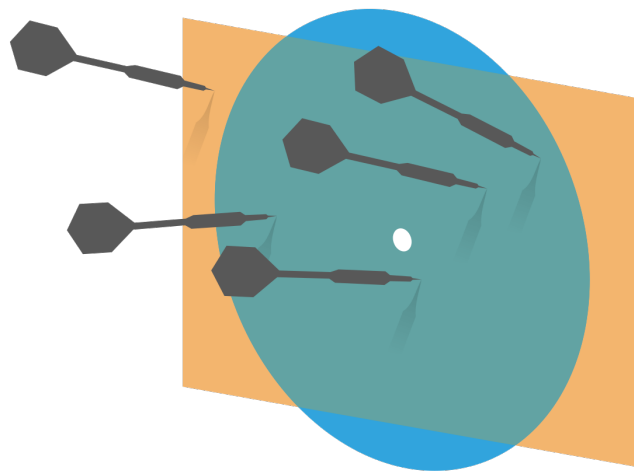


Probability and Statistics for Computer Science



Can we call e the
exciting e ?

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

Credit: wikipedia

Last time

- ✱ Important known discrete probability distributions
- ✱ Continuous Random Variable

Objectives

- ✱ Continuous RVs and their expected values
- ✱ Continuous uniform distribution
- ✱ Poisson distribution *Discrete*
- ✱ Exponential distribution *Cont!.*

How many empty slots?

Hashing N items to k slots ($N \geq k$) collisions are allowed, and will be handled by linked list. What is the expected number of empty slots?

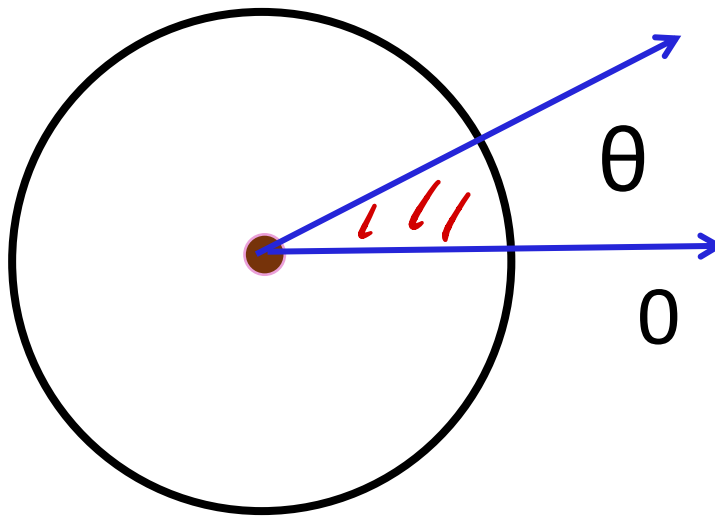
$$X_i = \begin{cases} 1 & \text{slot } i \text{ remains empty after hashing} \\ 0 & \text{otherwise} \end{cases}$$

$$P(X_i = 1) = ? \quad E[X_i] = ?$$

$$E[X_{Es}] = ?$$

Example of a continuous random variable

✱ The spinner



$$\theta \in (0, 2\pi]$$

$$\underline{P(a < \theta < b)}$$

✱ The sample space for all outcomes is not countable

~~$P(\theta = \theta_0)$ not defined.~~

Probability density function (pdf)

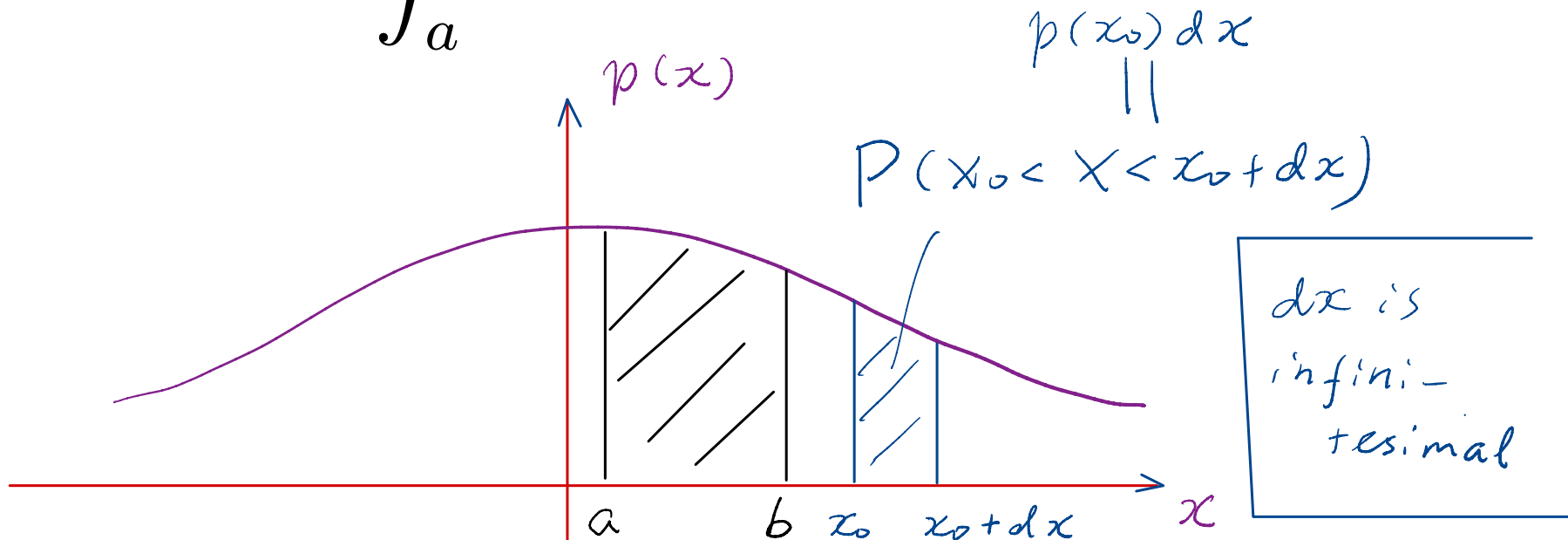
- ✱ For a continuous random variable X , the probability that $X=x$ is essentially zero for all (or most) x , so we can't define $P(X = x)$
- ✱ Instead, we define the **probability density function** (pdf) over an infinitesimally small interval dx , $p(x)dx = P(X \in [x, x + dx])$
- ✱ For $a < b$
$$\int_a^b p(x)dx = P(X \in [a, b])$$

Handwritten red notes:
 $p(x_0) = \lim_{dx \rightarrow 0} \frac{P(x_0 < X < x_0 + dx)}{dx}$

Probability density function (pdf)

✱ For $a < b$

$$\int_a^b p(x) dx = P(X \in [a, b])$$



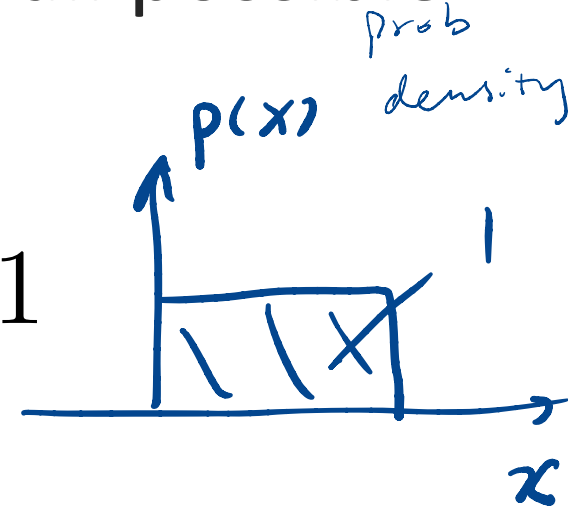
$$P(a < X < b) = \int_{-\infty}^b p(x) dx - \int_{-\infty}^a p(x) dx$$

$$P(X < a) = \int_{-\infty}^a p(x) dx$$

Properties of the probability density function

- ✱ $p(x)$ **resembles** the probability function of discrete random variables in that
 - ✱ $p(x) \geq 0$ for all x
 - ✱ The probability of X taking all possible values is 1.

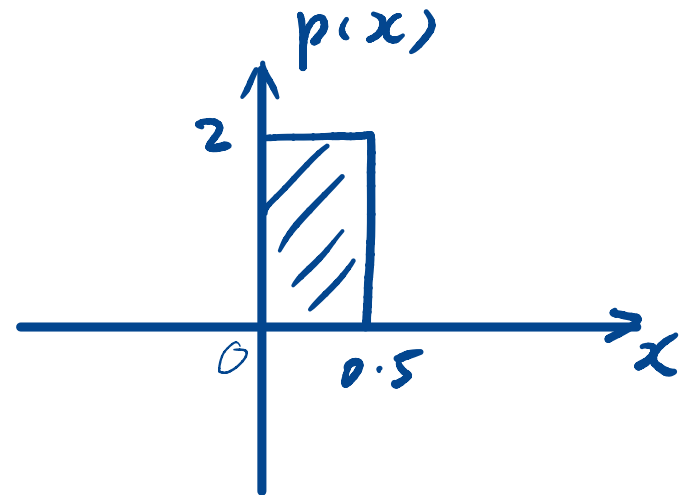
$$\int_{-\infty}^{\infty} p(x) dx = 1$$



Properties of the probability density function

- ✱ $p(x)$ **differs** from the probability distribution function for a discrete random variable in that
 - ✱ $p(x)$ is not the probability that $X = x$
 - ✱ $p(x)$ can exceed 1

$$\int_{-\infty}^{\infty} p(x) dx = 1$$

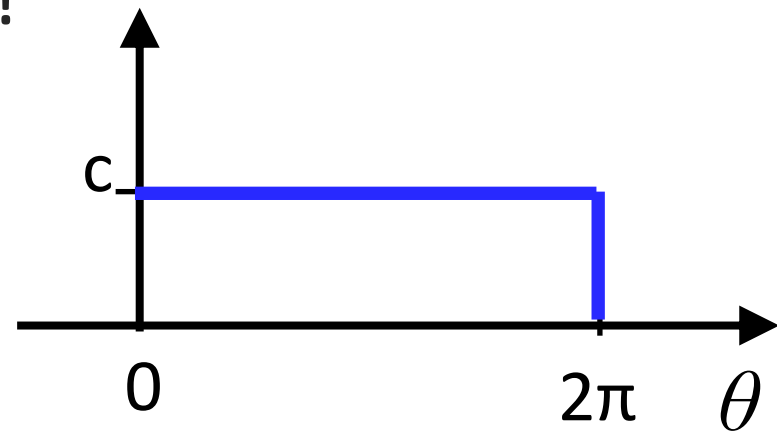


Example

Probability density function: spinner

- ✱ Suppose the spinner has equal chance stopping at any position. What's the pdf of the angle θ of the spin position?

$$p(\theta) = \begin{cases} c & \text{if } \theta \in (0, 2\pi] \\ 0 & \text{otherwise} \end{cases}$$



- ✱ For this function to be a pdf,

Then

$$\int_{-\infty}^{\infty} p(\theta) d\theta = 1$$

$$\int_0^{2\pi} c d\theta = 1$$

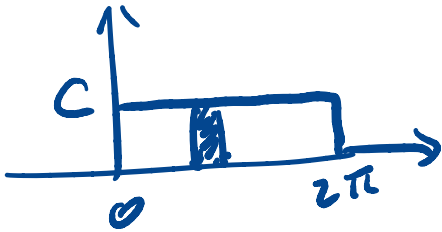
$$c = ?$$

$$2\pi \cdot c = 1 \\ c = \frac{1}{2\pi}$$

Probability density function: spinner

- ✱ What the probability that the spin angle θ is within $[\frac{\pi}{12}, \frac{\pi}{7}]$?

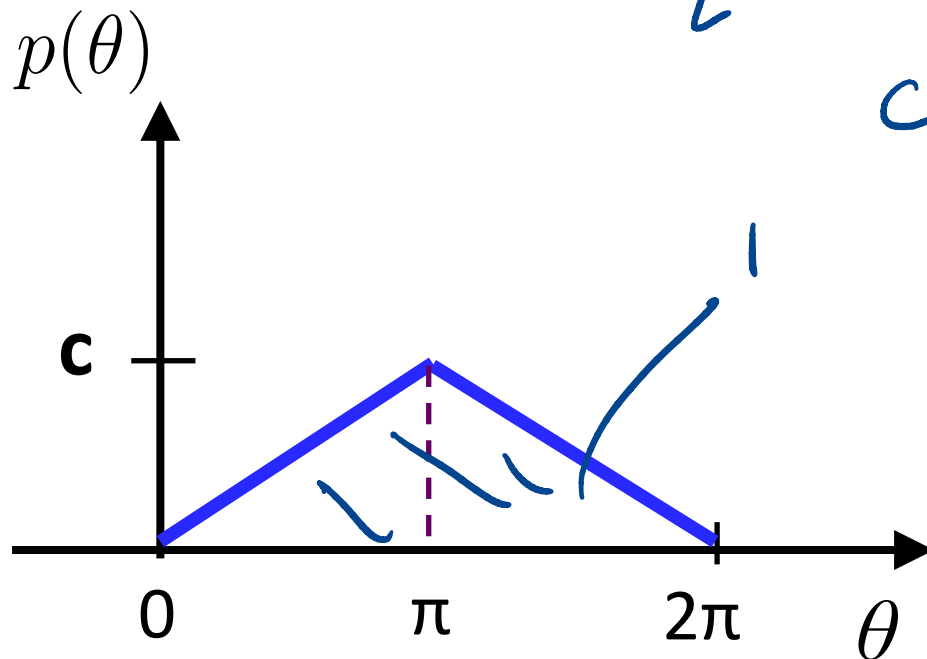
$$P\left(\theta \in \left[\frac{\pi}{12}, \frac{\pi}{7}\right]\right)$$



$$\begin{aligned} P(a < X < b) &= \int_a^b p(x) dx \\ &= \frac{1}{2\pi} \cdot x \Big|_{\frac{\pi}{12}}^{\frac{\pi}{7}} \\ &= \frac{1}{2} \left(\frac{1}{7} - \frac{1}{12} \right) \end{aligned}$$

Q: Probability density function:

- ✱ What is the constant c given the spin angle θ has the following pdf?



$$\frac{1}{2} c \cdot 2\pi = 1$$

$$c = \frac{1}{\pi}$$

A. 1

B. $1/\pi$

C. $2/\pi$

D. $4/\pi$

E. $1/2\pi$

Expectation of continuous variables

- Expected value of a continuous random variable X

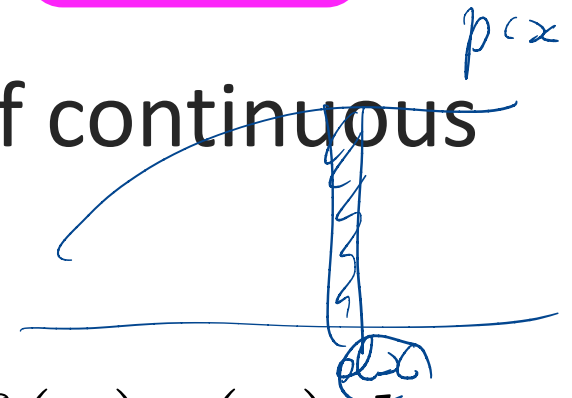
$$E[X] = \sum_x x p(x)$$

$$E[X] = \int_{-\infty}^{\infty} x p(x) dx$$

$p(X \in [x, x+dx])$
weight

- Expected value of function of continuous random variable $Y = f(X)$

$$E[Y] = E[f(X)] = \int_{-\infty}^{\infty} f(x) p(x) dx$$



Properties of expectation of *any* continuous random variables

- ✱ The linearity of expected value is true for continuous random variables.

$$\Sigma \longrightarrow \int \overset{E(ax + bY)}{=} a E[X] + b E[Y]$$

- ✱ And the other properties that we derived for variance and covariance also hold for continuous random variable

$$\text{var}[X] = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$
$$\text{cov}(X, Y) = E[(X - E[X])(Y - E[Y])] = E[XY] - E[X]E[Y]$$

Variance and covariance for continuous RVs

any

$$\begin{aligned} \text{var}[X] &= E[X^2] - (E[X])^2 \\ &= \int_{-\infty}^{\infty} x^2 p(x) dx - \left(\int_{-\infty}^{\infty} xp(x) dx \right)^2 \end{aligned}$$

$$\begin{aligned} \text{cov}(X, Y) &= E[XY] - E[X]E[Y] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xyp(x, y) dx dy - \int_{-\infty}^{\infty} xp(x) dx \cdot \int_{-\infty}^{\infty} yp(y) dy \end{aligned}$$

$E_x.$

Probability density function: spinner

✱ Given the probability density of the spin angle θ

$$p(\theta) = \begin{cases} \frac{1}{2\pi} & \text{if } \theta \in (0, 2\pi] \\ 0 & \text{otherwise} \end{cases}$$

✱ The expected value of spin angle is

$$E[\theta] = \int_{-\infty}^{\infty} \theta p(\theta) d\theta = \int_0^{2\pi} \theta \cdot \frac{1}{2\pi} d\theta = \frac{1}{2\pi} \cdot \frac{\theta^2}{2} \Big|_0^{2\pi} = \pi$$

Q.

✱ Suppose a continuous variable has pdf

$$p(x) = \begin{cases} 2(1-x) & x \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

What is $E[X]$?

A. $1/2$

B. $1/3$

C. $1/4$

D. 1

E. $2/3$

$$E[X] = \int_{-\infty}^{\infty} xp(x)dx$$

Cumulative distri. of a continuous RV

$$P(X \leq x) = \int_{-\infty}^x p(x) dx \rightarrow \text{CDF}(x)$$

$$\sum_{-\infty}^x p(x)$$

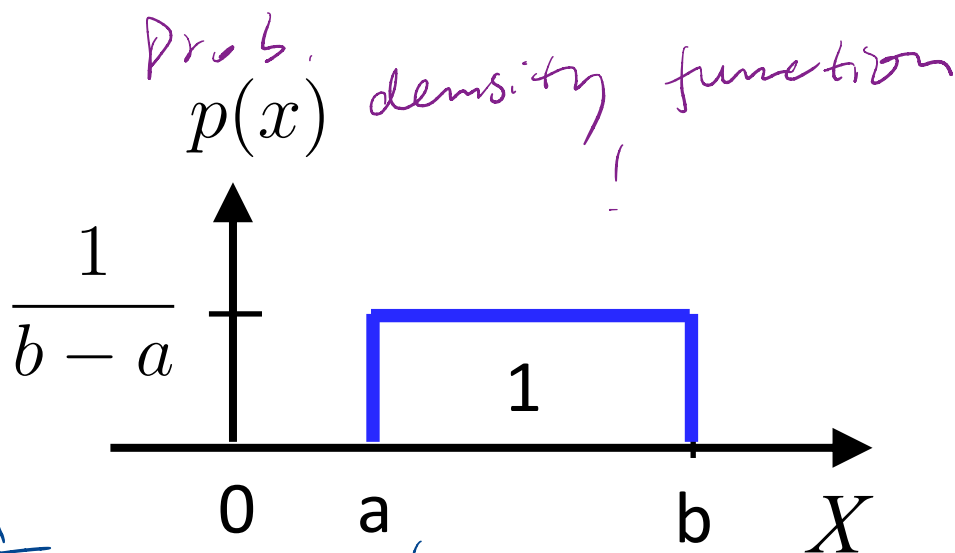
for Discrete RV.

$$P(X \leq x)$$

$$\text{CDF}(x) \text{ --- } F(x)$$

Continuous uniform distribution

- ✱ A continuous random variable X is uniform if

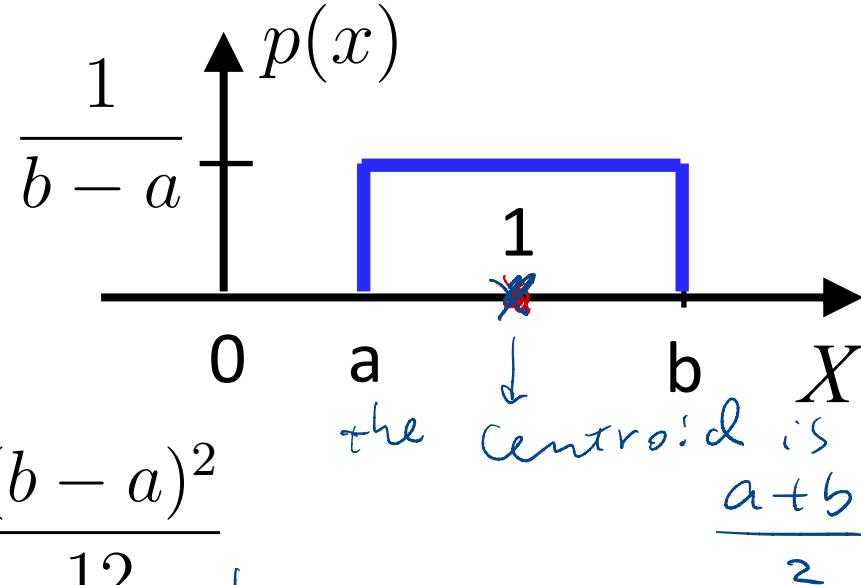


$$p(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0 & \text{ow} \end{cases}$$

Continuous uniform distribution

✱ A continuous random variable X is uniform if

$$p(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$



$$E[X] = \frac{a+b}{2} \quad \& \quad \text{var}[X] = \frac{(b-a)^2}{12}$$

$$E[X] = \int_{-\infty}^{\infty} x \cdot p(x) dx$$

$$= \int_a^b x \cdot \frac{1}{b-a} dx$$

$$= \frac{1}{b-a} \cdot \frac{x^2}{2} \Big|_a^b = \frac{a+b}{2}$$

$$\text{var}[X] = E[X^2] - (E[X])^2$$

$$= \int_a^b x^2 \cdot p(x) dx - \left[\frac{a+b}{2} \right]^2$$

$$= \int_a^b x^2 \cdot \frac{1}{b-a} dx - \left(\frac{a+b}{2} \right)^2$$

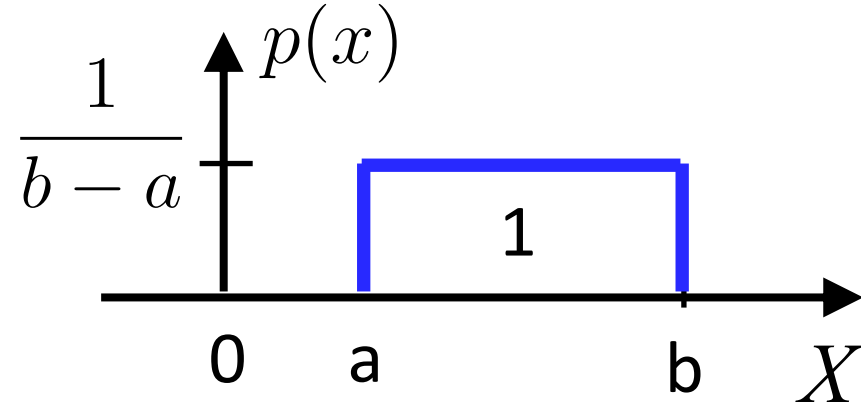
$$= \frac{1}{b-a} \left. \frac{x^3}{3} \right|_a^b - \left(\frac{a+b}{2} \right)^2$$

$$= \frac{(b-a)^2}{12}$$

Continuous uniform distribution

- ✱ A continuous random variable X is uniform if

$$p(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$



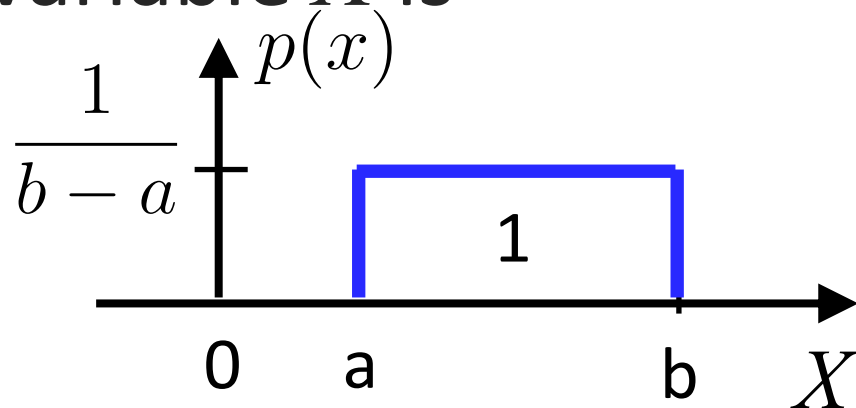
$$E[X] = \frac{a+b}{2} \quad \& \quad \text{var}[X] = \frac{(b-a)^2}{12}$$

- ✱ Examples: 1) A dart's position thrown on the target

Continuous uniform distribution

- ✱ A continuous random variable X is uniform if

$$p(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$



$$E[X] = \frac{a+b}{2} \quad \& \quad \text{var}[X] = \frac{(b-a)^2}{12}$$

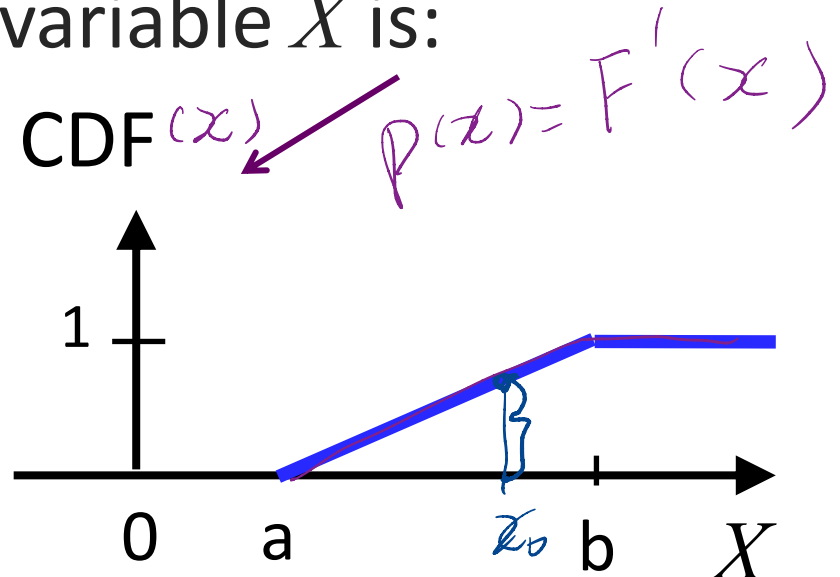
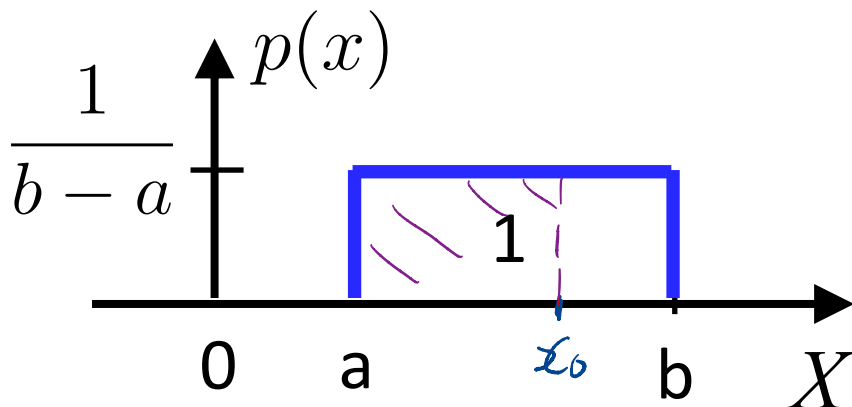
- ✱ Examples: 1) A dart's position thrown on the target 2) Often associated with **random sampling**

Cumulative distribution of continuous uniform distribution

✱ Cumulative distribution function (CDF)

$$P(X \leq x) = \int_{-\infty}^x p(x) dx = F(X=x) = F(x)$$

CDF of a uniform random variable X is:



What is the number?

$$e^x = \sum_{k=0}^{\infty} a_k x^k$$

$$a_k = ? \quad \frac{1}{k!}$$

What is the number?

$$e^x = \sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

$$a_k = ?$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$

$$e^0 = 1$$

$$(e^x)' = e^x \checkmark$$

Motivation for a model called Poisson Distribution

- ✱ What's the probability of the **number of incoming customers (k)** in an hour?
- ✱ It's widely applicable in physics and engineering both for modeling of time and space.



Simeon D. Poisson Credit: wikipedia
(1781-1840)

De Groot
Pg 287-288

Poisson Distribution

Discrete!!

- ✱ A discrete random variable X is called **Poisson** with intensity λ ($\lambda > 0$) if

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

for integer $k \geq 0$



Simeon D. Poisson
(1781-1840)

λ is the average rate of the event's occurrence

Poisson Distribution

✱ **Poisson** distribution is a valid pdf for

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

$$\begin{aligned} \sum_{k=0}^{\infty} P(X=k) &= \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} e^{\lambda} = e^0 = 1 \end{aligned}$$

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

for integer $k \geq 0$



Simeon D. Poisson
(1781-1840)

λ is the average rate of the event's occurrence

Poisson Distribution

✱ **Poisson** distribution is a valid pdf for

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda} \Rightarrow \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = 1$$



Simeon D. Poisson
(1781-1840)

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

for integer $k \geq 0$

λ is the average rate of the event's occurrence

Expectations of Poisson Distribution

✱ The expected value and the variance are wonderfully the same! That is λ

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

$E[X] = \sum_{k=0}^{\infty} k P(k)$ $N \cdot p$

for integer $k \geq 0$

$p \rightarrow 0$
 $NP(1-p)$



Simeon D. Poisson
(1781-1840)

$$E[X] = \lambda$$

$$\text{var}[X] = \lambda$$

$$E[X] = \sum_x x P(x)$$

$$= \sum_k k P(X=k)$$

$$= \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!}$$

$$= \sum_{k=1}^{\infty} \cancel{k} \frac{e^{-\lambda} \lambda^{k-1} \cdot \lambda}{\cancel{k} (k-1)!}$$

$$= \sum_{k=1}^{\infty} \lambda \frac{e^{-\lambda} \lambda^{k-1}}{(k-1)!}$$

$$= \lambda$$

$$\text{var}[X] = E[X^2] - (E[X])^2$$

$$E[X^2] = \sum x^2 p(x)$$

$$= \sum_{k=0}^{\infty} k^2 \frac{e^{-\lambda} \lambda^k}{k!}$$

$$= \sum_{k=2}^{\infty} \frac{k^2}{k(k-1)} \frac{e^{-\lambda} \lambda^{k-2} \cdot \lambda^2}{(k-2)!}$$

$$+ \sum_{k=0}^1 k^2 \frac{e^{-\lambda} \lambda^k}{k!}$$

$$= \lambda + \lambda^2$$

$$\text{var}[X] = E[X^2] - (E[X])^2 = \lambda$$

$$1 + \frac{1}{k-1}$$

Examples of Poisson Distribution

- ✱ How many calls does a call center get in an hour?
- ✱ How many mutations occur per 100k nucleotides in a DNA strand?
- ✱ How many **independent** incidents occur in an interval?

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

for integer $k \geq 0$

Poisson Distribution: call center

✱ If a call center receives 10 calls per hour on average, what is the probability that it receives 15 calls in a given hour?

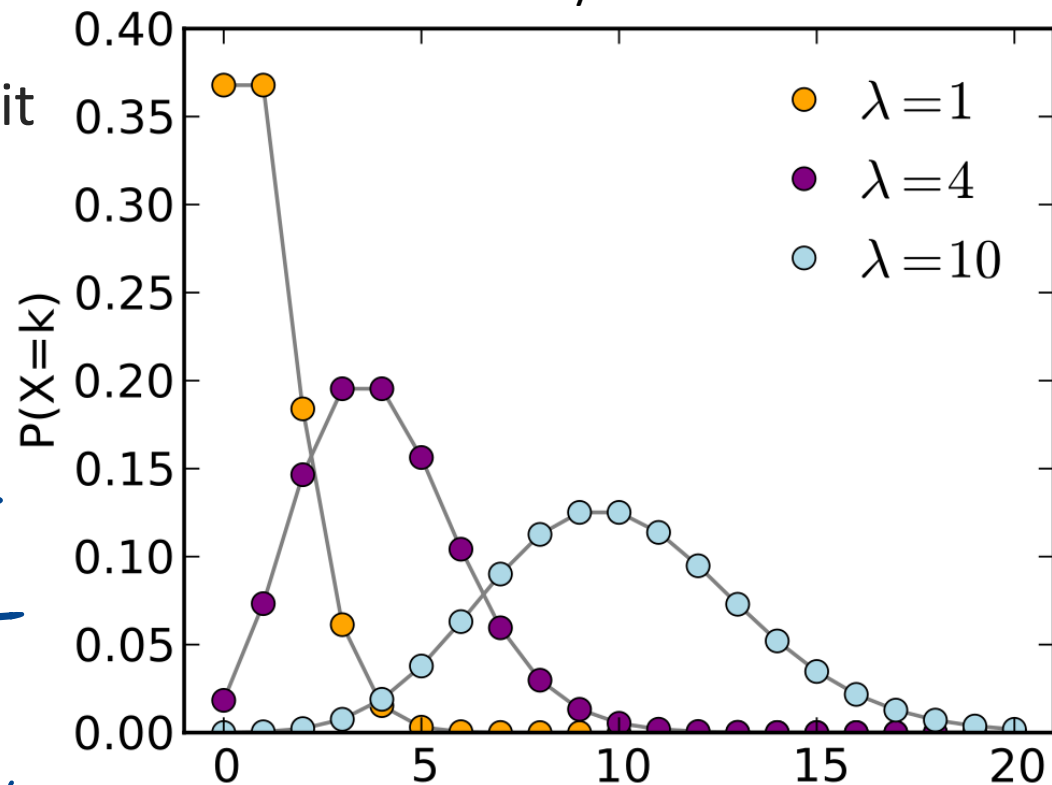
✱ What is λ here? 10

✱ What is $P(k=15)$? $e^{-10} \cdot 10^{15} / 15!$

$$P(k) = \frac{e^{-\lambda} \lambda^k}{k!} = \frac{e^{-10} 10^{15}}{15!}$$

$$P(k < 10) = \sum_{k=0}^9 P(k)$$

*The lines are only to show the trend



$$P(k > 10) = 1 - P(k \leq 10)$$

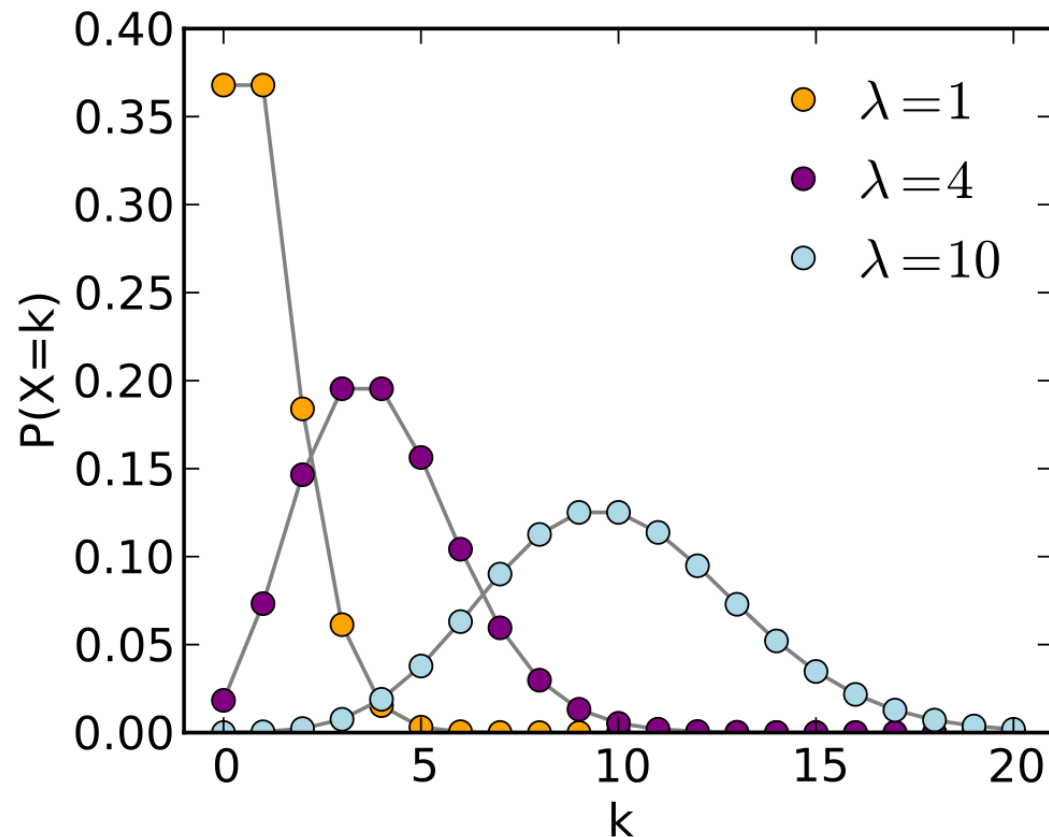
Credit: wikipedia

Q. Poisson Distribution: call center

If a call center receives 4 calls per hour on average.

What is probability the center receives 0 calls in an hour?

- ☒ A. e^{-4}
- B. 0.5
- C. 0.05

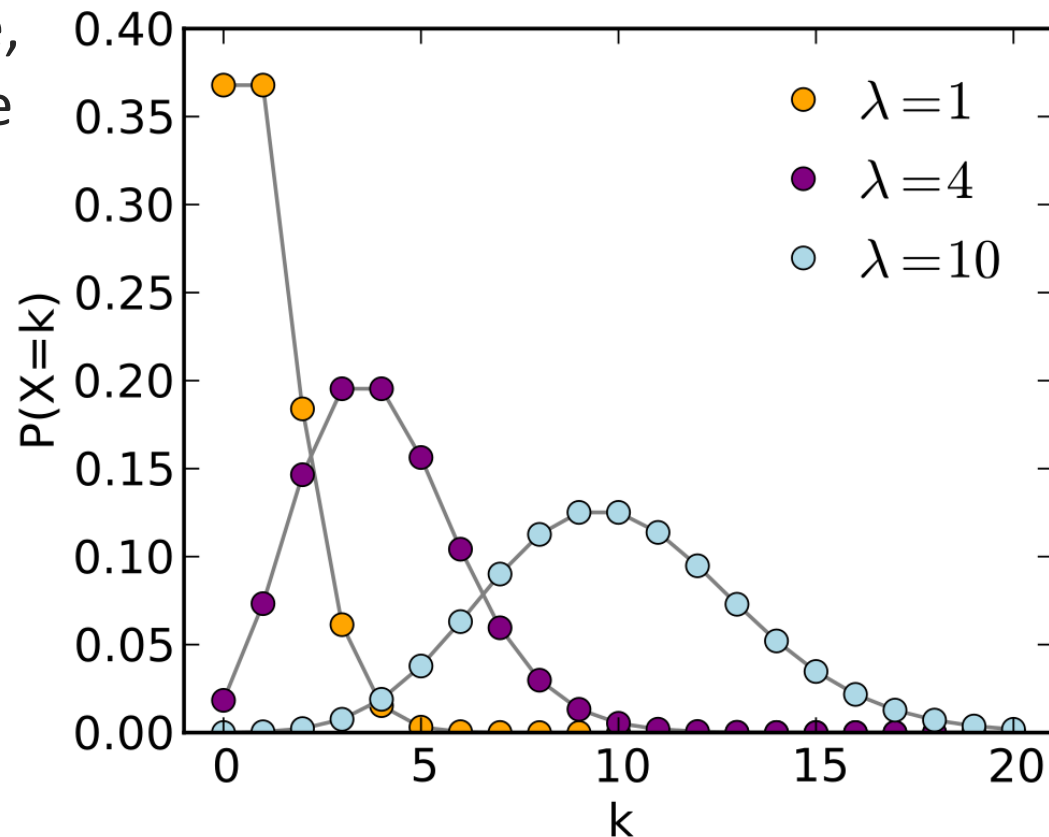


Credit: wikipedia

Q. Poisson Distribution: call center

- ✱ Given a call center receives 10 calls per hour on average, what is the intensity λ of the distribution for calls in **Two** hours?

$$\lambda = 20$$

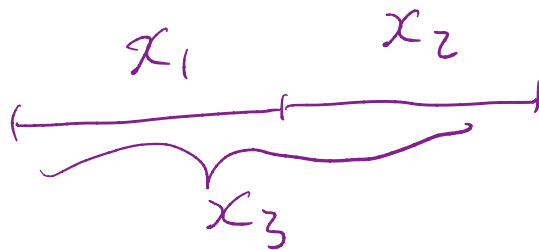


Credit: wikipedia

Q. Poisson Distribution: call center

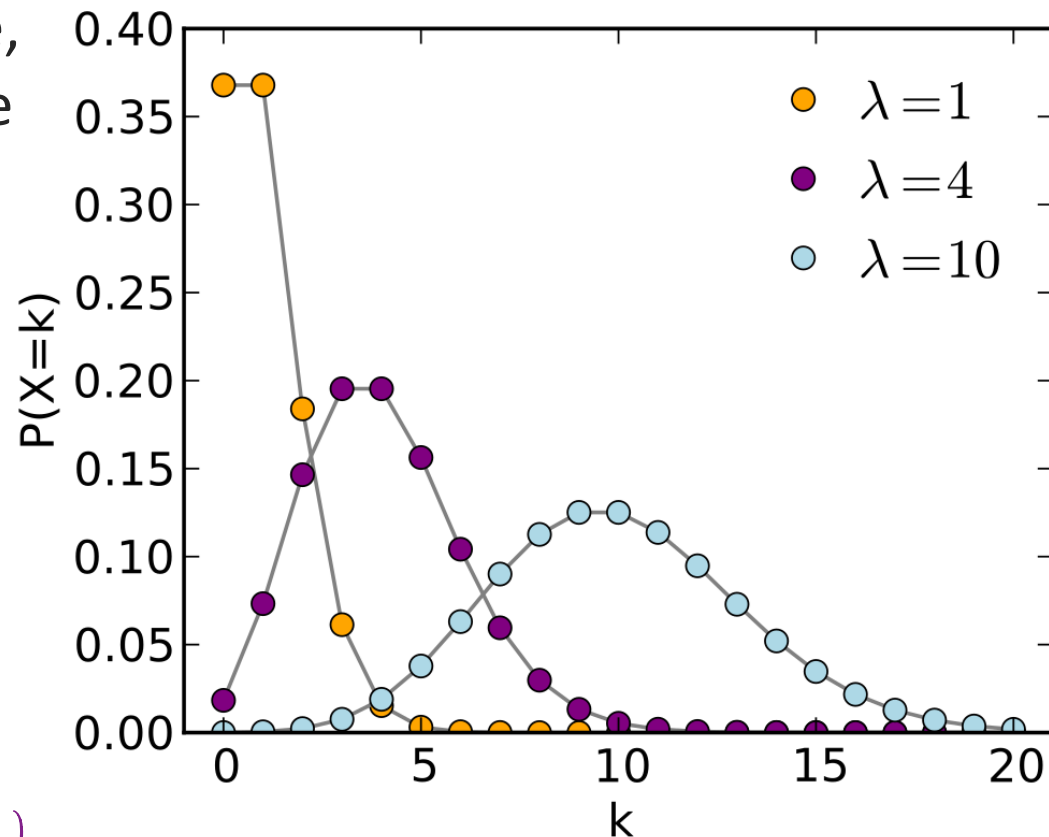
✱ Given a call center receives 10 calls per hour on average, what is the intensity λ of the distribution for calls in **Two** hours? $\lambda = 20$

Ans:



$$X_3 = X_1 + X_2$$

$$\begin{aligned} E[X_3] &= E[X_1] + E[X_2] \\ &= \lambda_1 + \lambda_2 = 2\lambda_1 = 20 \end{aligned}$$



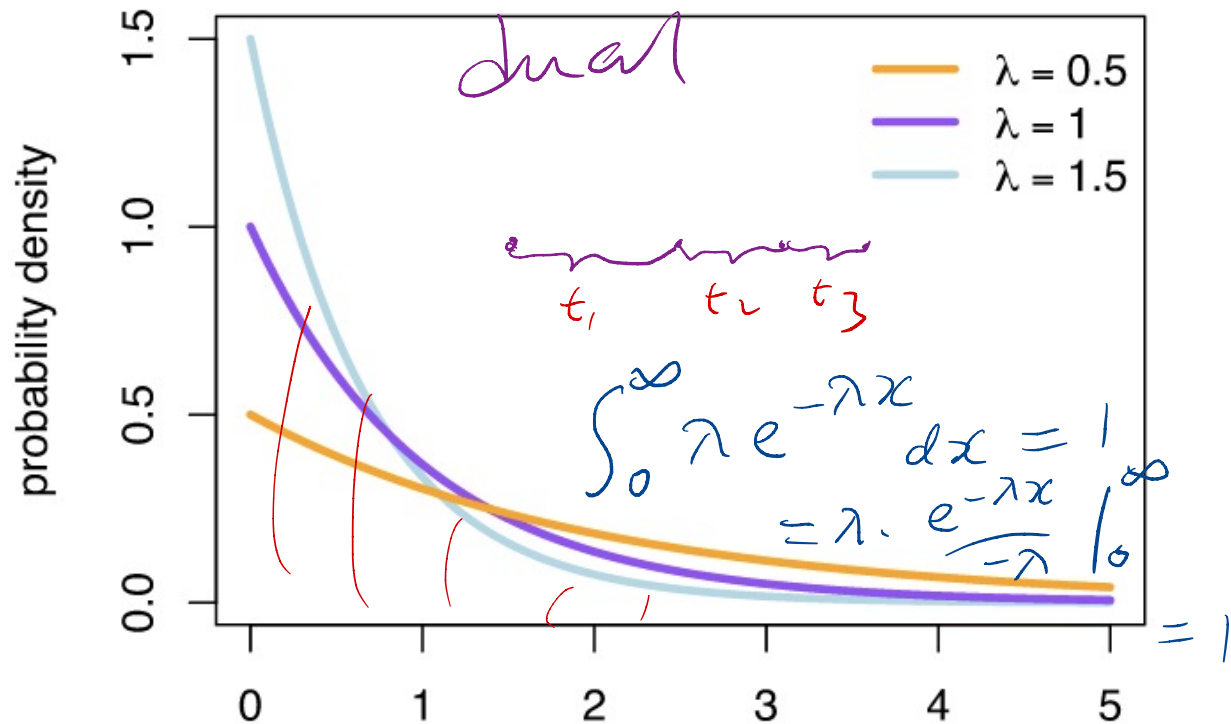
Credit: wikipedia

Exponential distribution *Continuous*

- ✱ Common Model for waiting time

- ✱ Associated with the Poisson distribution with the same λ

$$p(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$



Expectations of Exponential distribution

- ✱ A continuous random variable X is exponential if it represents the “time” until next incident in a Poisson distribution with intensity λ .

$$p(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

$$E[X] = \frac{1}{\lambda} \quad \& \quad \text{var}[X] = \frac{1}{\lambda^2}$$

$E[X]$ of Exponential Distr:

$$E[X] = \int_{-\infty}^{\infty} x \cdot p(x) dx$$

$$= \int_0^{\infty} x \cdot \lambda e^{-\lambda x} dx$$

$$= -x e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx \quad \leftarrow \text{by parts}$$

$$= 0 - 0 + \frac{e^{-\lambda x}}{-\lambda} \Big|_0^{\infty}$$

$$= \frac{1}{\lambda}$$

$$\begin{aligned} (x \cdot e^{-\lambda x})' &= x \cdot (-\lambda) e^{-\lambda x} + e^{-\lambda x} \\ &\Rightarrow (-x \lambda e^{-\lambda x})' \\ &= x \lambda e^{-\lambda x} - e^{-\lambda x} \end{aligned}$$

Example of exponential distribution

- ✱ How long will it take until the next call to be received by a call center? Suppose it's a random variable T . If the number of incoming call is a Poisson distribution with intensity $\lambda = 20$ in ~~an hour~~. What is the expected time for T ?

$$\lambda = 20$$

$$\lambda^* = \frac{20}{2}$$

$$E[X] = \frac{1}{\lambda^*} = \frac{1}{10} \text{ (hr)}$$

$$E[X] = \frac{1}{\lambda} = \frac{1}{20} \text{ (hr)}$$

↓
exponential

$$= 3 \text{ mins.}$$

Example of exponential distribution

- ✱ How long will it take until the next call to be received by a call center? Suppose it's a random variable T . If the number of incoming call is a Poisson distribution with intensity $\lambda = 20$ in ~~an hour~~. What is the expected time for T ? *2 hours*

$$E[T] = \frac{1}{\lambda} = \left(\frac{1}{20} \right) (2h) = 3 \text{ mins}$$
$$= \frac{1}{10} h = 6 \text{ mins}$$

Q:

✱ A store has a number of customers coming on Sat. that can be modeled as a Poisson distribution. In order to measure the average rate of customers in the day, the staff recorded the time between the arrival of customers, can he reach the same goal?

☒ A. Yes B. No

Assignments

- ✱ Work week6 module
- ✱ Next time: Normal distribution, CLT

Additional References

- ✱ Charles M. Grinstead and J. Laurie Snell
"Introduction to Probability"
- ✱ Morris H. Degroot and Mark J. Schervish
"Probability and Statistics"

Qs for discrete distributions



Q.

✱ A store staff mixed their fuji  and gala  apples and they were individually wrapped, so they are indistinguishable. Given there are 70% of fuji, if I want to know what is the probability I get 7 fuji in 20 apples? What is the distribution I should use?

A. Bernoulli

B. Binomial

C. Geometric

D. Poisson

E. Uniform

Q.

- ✱ A store staff mixed their fuji  and gala  apples and they were individually wrapped, so they are indistinguishable. Given there are 70% of fuji, if I want to know what is the probability I get 7 fuji in 20 apples? What is the distribution I should use? **What is the probability?**

Q.

✱ A store staff mixed their fuji  and gala  apples and they were individually wrapped, so they are indistinguishable. Given there are 70% of fuji, if I want to know the probability of picking the first gala on the 7th time (I can put back after each pick). What is the distribution I should use?

A. Bernoulli

B. Binomial

C. Geometric

D. Poisson

E. Uniform

Q.

- ✱ A store staff mixed their fuji  and gala  apples and they were individually wrapped, so they are indistinguishable. Given there are 70% of fuji, if I want to know the probability of picking the first gala on the 7th time (I can put back after one pick). **What's the probability?**

Q.

- ✱ A store staff mixed their fuji  and gala  apples and they were individually wrapped, so they are indistinguishable. Given there are 70% of fuji, **what's the average times of picking to get the first gala?**

See you next time

*See
You!*

